

CompTIA.

# A+ Core 2

Exam 220-1202

A+ 220-1202

# Module 17



## Supporting Other OS

Prepare for A+ Core 2 by:

- Identifying features of Linux
- Identifying features of macOS

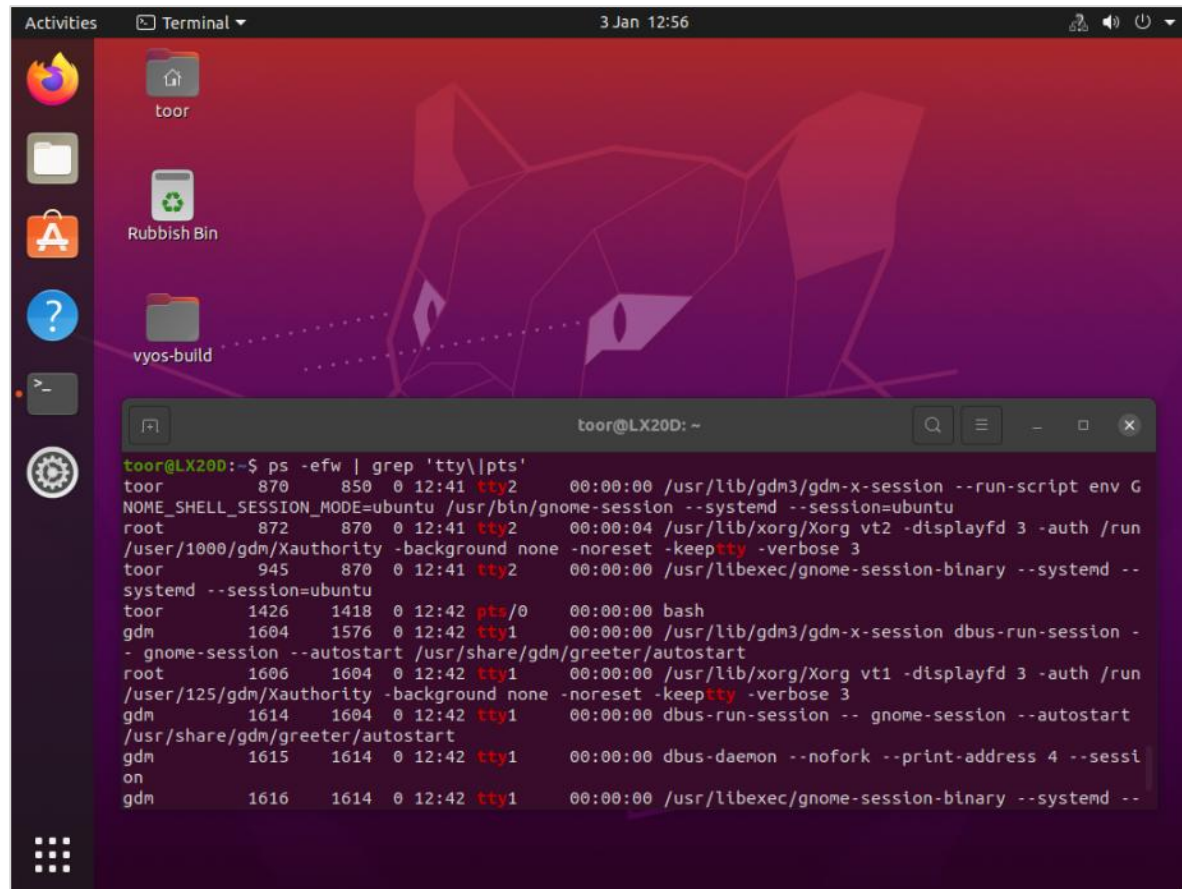
# Lesson 17.1

## Linux Features

# Shells, Terminals, and Consoles (Slide 1 of 2)

- Bootloader
- Shell and Terminal
  - Shell provides command line interface for user
  - Bash, zsh, ksh are popular shells
  - Terminal application connects to shell interpreter
    - stdin, stdout, stderr
- Desktop environment
  - GUI for user interaction
- Console switching
  - CTRL+ALT+Fx

# Shells, Terminals, and Consoles (Slide 2 of 2)



# Command Interface

Command

- Command to be executed

Options

- Modification of the command

Arguments

- Values that operate the command

Pipes

- Redirection of one commands output as the input to second command

String multiple commands

- Use semicolon (;) to string commands together

Case sensitivity

Help options

Text Editor

- Vi vim nano

# Navigation Commands (Slide 1 of 3)



No drive letters in Linux

Everything is a directory or sub directory of the root of the file system



pwd

Present working directory  
Your location currently within the file system



cd

Change directory



ls

List contents of directory

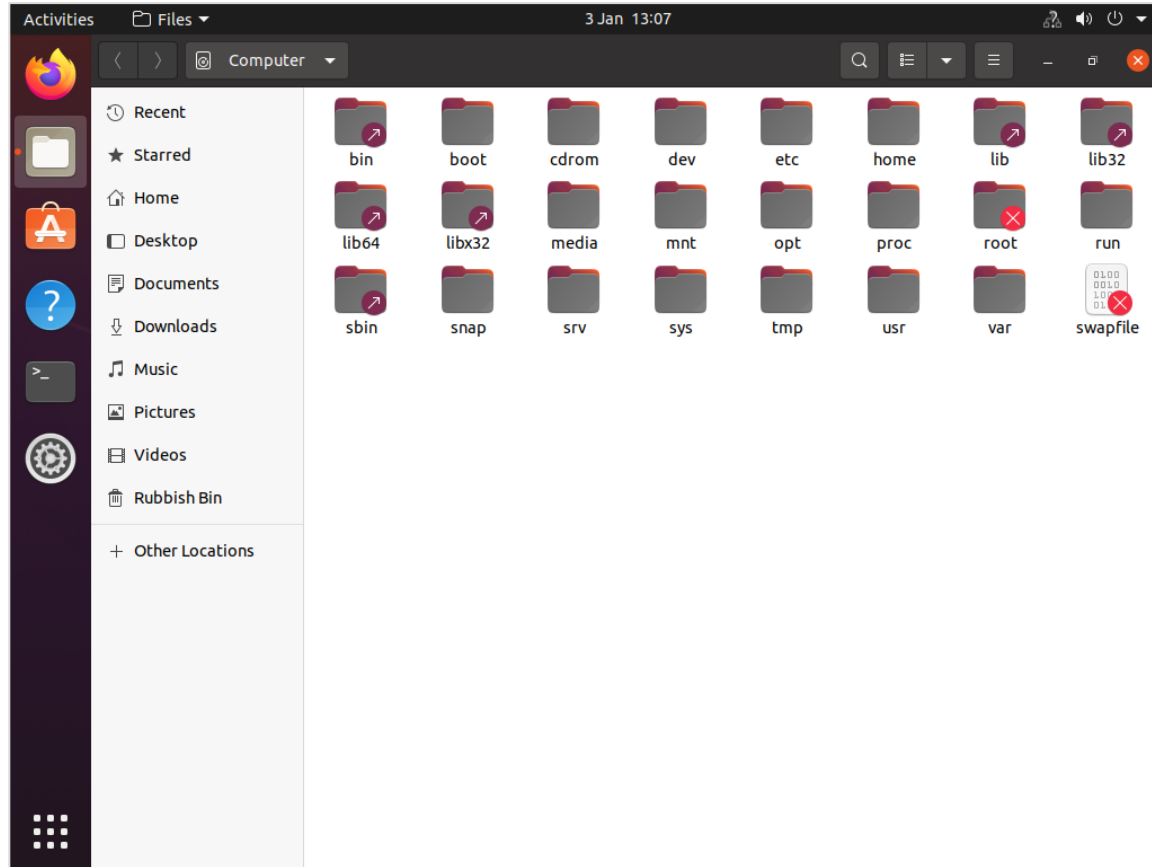


cat

Display the contents of a file



# Navigation Commands (Slide 2 of 3)



# Navigation Commands (Slide 3 of 3)

```
tj@TJ-VM:~/Documents/lessons$
```

# Search Commands (Slide 1 of 3)

find to search for file by name, owner, size, etc.

grep to search for strings within a file

- -i option will ignore case sensitivity

Metacharacters and escaping

- \* wildcard character
- Backslash \
- Single and double quotes ' ' or " "

# Search Commands (Slide 2 of 3)

```
tj@TJ-VM:/var/log$
```

# Search Commands (Slide 3 of 3)

```
tj@TJ-VM:/var/log$
```

# Filesystem Management

## mount

- Logical connection of file system to the OS

## /etc/fstab

- File system information file dictating file system mount points

## fsck (File system check)

- Check and repair utility

# File Management Commands

- cp
  - copy
- mv
  - move
- rm
  - remove
- df
  - disk space free statistics
- du
  - disk space used statistics

# User Account Management

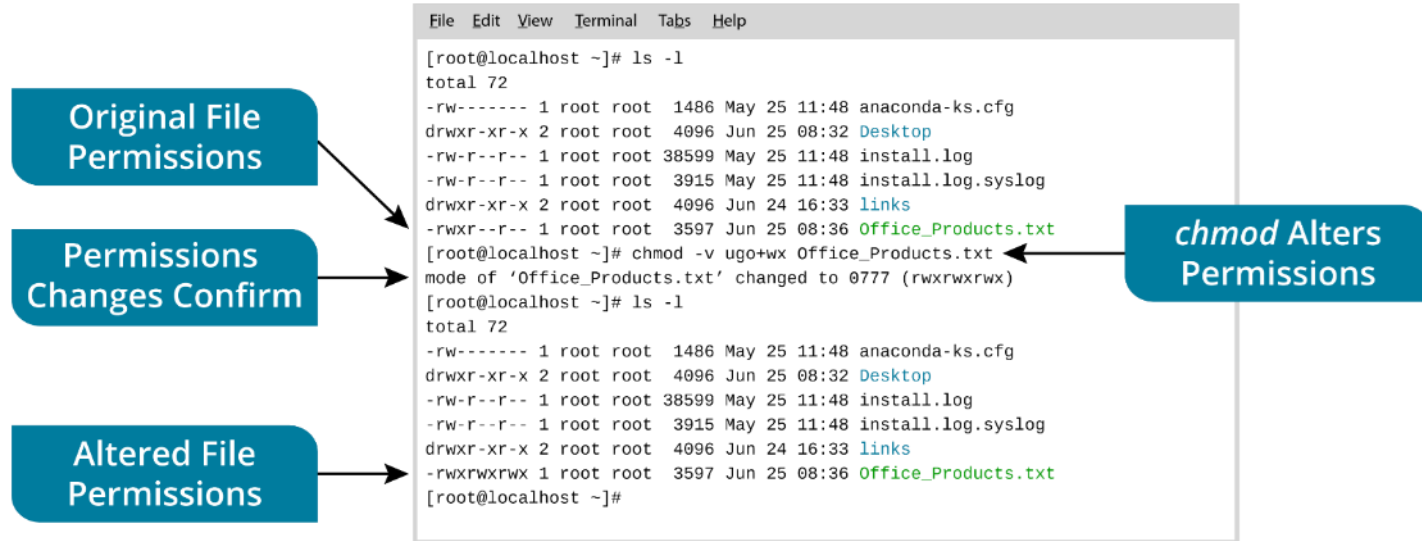
- su
  - switch user
- sudo
  - Run specified commands with elevated permissions
  - Account must be listed in /etc/sudoers file
- /etc/passwd - The /etc/passwd file is a **plain-text database** housing fundamental user information
- /etc/shadow - For **heightened security**, critical user authentication information, **particularly hashed passwords**, resides in the /etc/shadow file, accessible exclusively to privileged users
- useradd, usermod, userdel
- groupadd, groupmod, groupdel



# File Permissions Commands (Slide 1 of 2)

- Read, Write, Execute (rwx)
- Symbolic mode using rwx designations or - for denied permission
- Octal mode
  - Read (4), Write (2), Execute (1)
- `chmod`
  - Change or modify permissions
- `chown`
  - Change ownership of file/directory

# File Permissions Commands (slide 2 of 2)



## Lab Activity

- Lab: File Management  
(available in CertMaster Learn and CertMaster Perform)
- Lab: Manage Linux File Ownership  
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Report Linux Configuration Settings  
(available in CertMaster Perform and CertMaster Labs)

## Lesson Review: Linux Features

- Shells, Terminals, and Consoles
- Command Interface
- Navigation Commands
- Search Commands
- Filesystem Management
- File Management Commands
- User Account Management
- File Permissions Commands

# Lesson 17.2

## Package and Network Management

# Package Management Commands (slide 1 of 2)

- **apt** Debian based distros (command-line utility for managing software (installing, updating, removing packages on Debian-based Linux distributions like Ubuntu and Mint)
  - apt update
  - apt upgrade
  - apt install
- **DNF(yum)** Red Hat distros (install, update, remove, and manage software packages and their dependencies automatically)
  - dnf check-update
  - dnf update
  - dnf upgrade
  - dnf install or dnf remove
- Updates are retrieved from repositories (repos)

# Package Management Commands (slide 2 of 2)

```
toor@LX20D:/home$ cat /etc/apt/sources.list
#deb cdrom:[Ubuntu 20.04.2.0 LTS _Focal Fossa_ - Release amd64 (20210209.1)]/ focal main restricted

# See http://help.ubuntu.com/community/UpgradeNotes for how to upgrade to
# newer versions of the distribution.
deb http://gb.archive.ubuntu.com/ubuntu/ focal main restricted
# deb-src http://gb.archive.ubuntu.com/ubuntu/ focal main restricted

## Major bug fix updates produced after the final release of the
## distribution.
deb http://gb.archive.ubuntu.com/ubuntu/ focal-updates main restricted
# deb-src http://gb.archive.ubuntu.com/ubuntu/ focal-updates main restricted

## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team. Also, please note that software in universe WILL NOT receive any
## review or updates from the Ubuntu security team.
deb http://gb.archive.ubuntu.com/ubuntu/ focal universe
# deb-src http://gb.archive.ubuntu.com/ubuntu/ focal universe
deb http://gb.archive.ubuntu.com/ubuntu/ focal-updates universe
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## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team, and may not be under a free licence. Please satisfy yourself as to
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deb http://gb.archive.ubuntu.com/ubuntu/ focal multiverse
# deb-src http://gb.archive.ubuntu.com/ubuntu/ focal multiverse
deb http://gb.archive.ubuntu.com/ubuntu/ focal-updates multiverse
# deb-src http://gb.archive.ubuntu.com/ubuntu/ focal-updates multiverse
```

# Process Monitoring Commands (Slide 1 of 3)

- `ps`
  - Display process table
- `top`
  - Display all running processes in real time statuses
- `systemd`
  - Init system and service manager in Linux
- `systemctl`
  - Used with `systemd` to start, stop, enable, and disable services



# Process Monitoring Commands (Slide 2 of 3)

The image shows a terminal window with the command `ps -e` executed. The output is a table of system processes. Annotations with arrows point to specific columns: 'Process ID' points to the PID column, 'Controlling terminal' points to the TTY column, 'CPU time' points to the TIME column, and 'Process command' points to the CMD column.

| PID | TTY | CPU time | CMD           |
|-----|-----|----------|---------------|
| 1   | ?   | 00:01:42 | systemd       |
| 2   | ?   | 00:00:00 | kthreadd      |
| 3   | ?   | 00:00:02 | ksoftirqd/0   |
| 5   | ?   | 00:00:00 | kworker/0:0H  |
| 7   | ?   | 00:00:02 | migration/0   |
| 8   | ?   | 00:00:00 | rcu_bh        |
| 9   | ?   | 00:05:55 | rcu_sched     |
| 10  | ?   | 00:00:00 | lru-add-drain |

# Process Monitoring Commands (Slide 3 of 3)

Total system  
details

```
root@server01:~  
top - 15:39:21 up 5 days, 20:58, 2 users, load average: 0.08, 0.07, 0.1  
Tasks: 291 total, 1 running, 290 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 4.0 us, 4.5 sy, 0.0 ni, 91.4 id, 0.0 wa, 0.0 hi, 0.0 si,  
KiB Mem : 7915376 total, 3992396 free, 1158144 used, 2764836 buff/cac  
KiB Swap: 8126460 total, 8126460 free, 0 used. 6022044 avail Me
```

System details  
by process

| PID          | USER        | PR        | NI       | VIRT          | RES         | SHR         | S        | %CPU       | %MEM       | TIME+          |
|--------------|-------------|-----------|----------|---------------|-------------|-------------|----------|------------|------------|----------------|
| 884          | polkitd     | 20        | 0        | 650036        | 19012       | 5432        | S        | 6.6        | 0.2        | 488:39.58      |
| 882          | dbus        | 20        | 0        | 71224         | 4700        | 1932        | S        | 2.0        | 0.1        | 167:55.82      |
| 2580         | student+    | 20        | 0        | 4063464       | 261620      | 73944       | S        | 1.7        | 3.3        | 16:59.89       |
| 885          | root        | 20        | 0        | 396304        | 6120        | 3196        | S        | 1.3        | 0.1        | 107:20.79      |
| <b>30994</b> | <b>root</b> | <b>20</b> | <b>0</b> | <b>162112</b> | <b>2408</b> | <b>1580</b> | <b>R</b> | <b>1.3</b> | <b>0.0</b> | <b>0:00.08</b> |
| 1921         | root        | 20        | 0        | 353532        | 90912       | 51112       | S        | 1.0        | 1.1        | 6:45.79        |
| 3504         | student+    | 20        | 0        | 769132        | 34712       | 17092       | S        | 1.0        | 0.4        | 0:46.50        |
| 1            | root        | 20        | 0        | 194068        | 7168        | 4192        | S        | 0.3        | 0.1        | 1:42.31        |
| 890          | root        | 20        | 0        | 13216         | 836         | 628         | S        | 0.3        | 0.0        | 0:16.22        |
| 2751         | student+    | 20        | 0        | 611428        | 6932        | 5188        | S        | 0.3        | 0.1        | 42:20.70       |

# Network Management Commands (Slide 1 of 2)

- ip
  - ip addr
- /etc/hosts
  - Host name to IP mapping locally; queried before DNS
- /etc/resolv.conf
  - DNS server information
- ping
  - Tests connectivity

# Network Management Commands (Slide 2 of 2)

- dig
  - DNS query through CLI
- curl
  - Transfer data from one system to another
- traceroute
  - Network diagnostic tool

# Backup and Scheduling Commands

- cron job
  - Task scheduler for Linux
- crontab
  - List of scheduled jobs
- Syntax for scheduling jobs
  - *mm hh dd MM*
  - *weekday*
  - *command*

## Lab Activity

- Lab: Informational and Network Tools  
(available in CertMaster Learn and CertMaster Perform)
- Lab: Configure Linux  
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Manage Linux Apps and Files  
(available in CertMaster Perform and CertMaster Labs)

## Lesson Review: Package and Network Management

- Package Management Commands
- Process Monitoring Commands
- Network Management Commands
- Backup and Scheduling Commands

# Lesson 17.3

## macOS Features



# Interface Features (Slide 1 of 3)

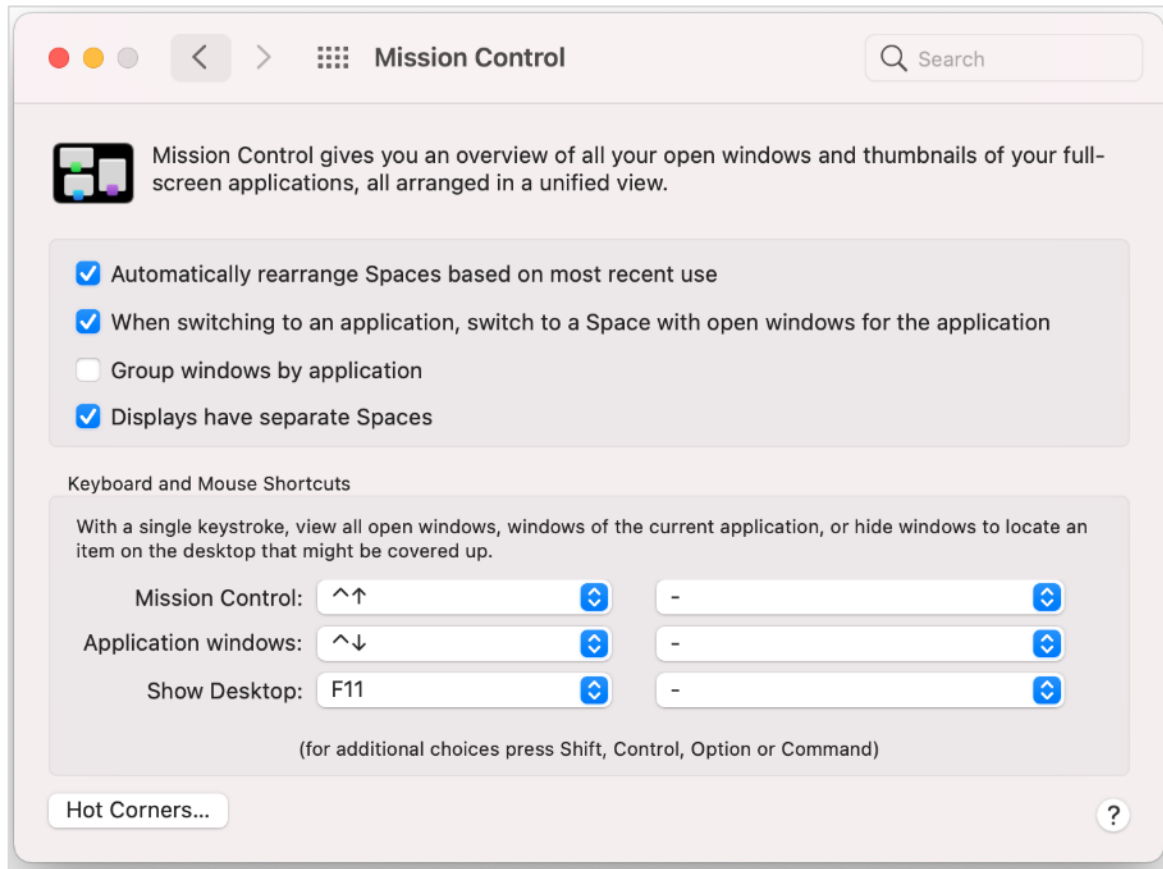
- Menu bar
  - Top of screen
- Dock
  - Bottom of screen with one click access to apps and files
- Spotlight Search searches content on system
  - COMMAND+SPACE
- Terminal
  - CLI for macOS
- Mission Control
  - Multiple desktop displays
  - F3 key to activate

# Interface Features (Slide 2 of 3)



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# Interface Features (Slide 3 of 3)

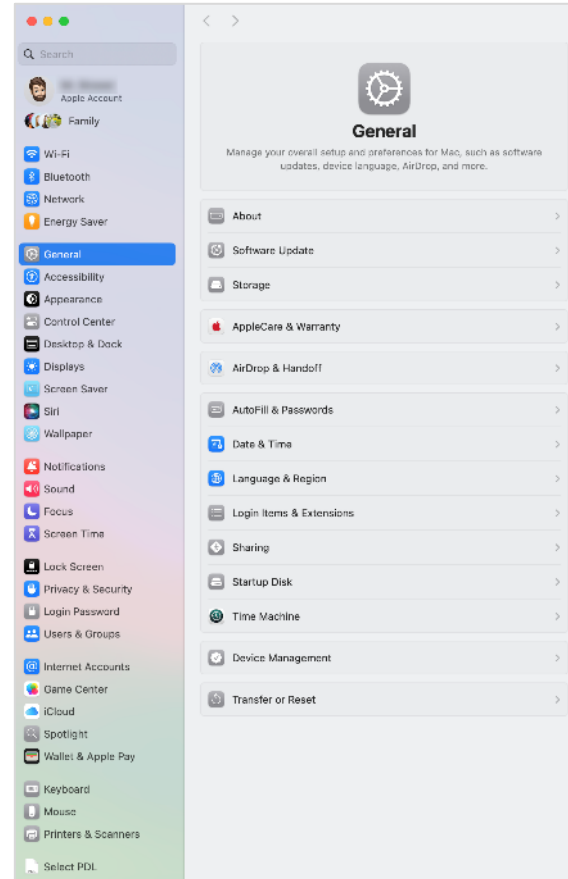


# System Folders and Finder

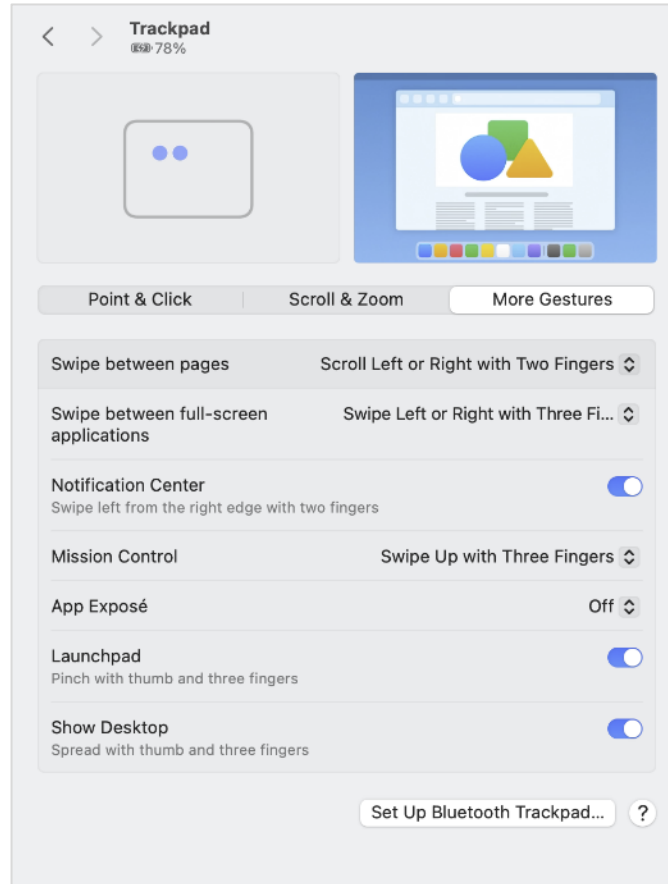
- Everything stems from root of directory
  - /Applications all applications installed
  - /Library system-wide resources and settings
  - /System core system files and resources for operation
  - /Users user account data
  - /Users/Library hidden folder for each user's home directory
- Finder
  - File Explorer for macOS

# System Settings (Slide 1 of 3)

- Settings, network options, and configurations
- Apple keyboard differences
  - COMMAND- CTRL on Windows
  - OPTION- ALT key
  - CONTROL
- Displays
  - Scaling and display settings
- Accessibility
  - Vision and sound

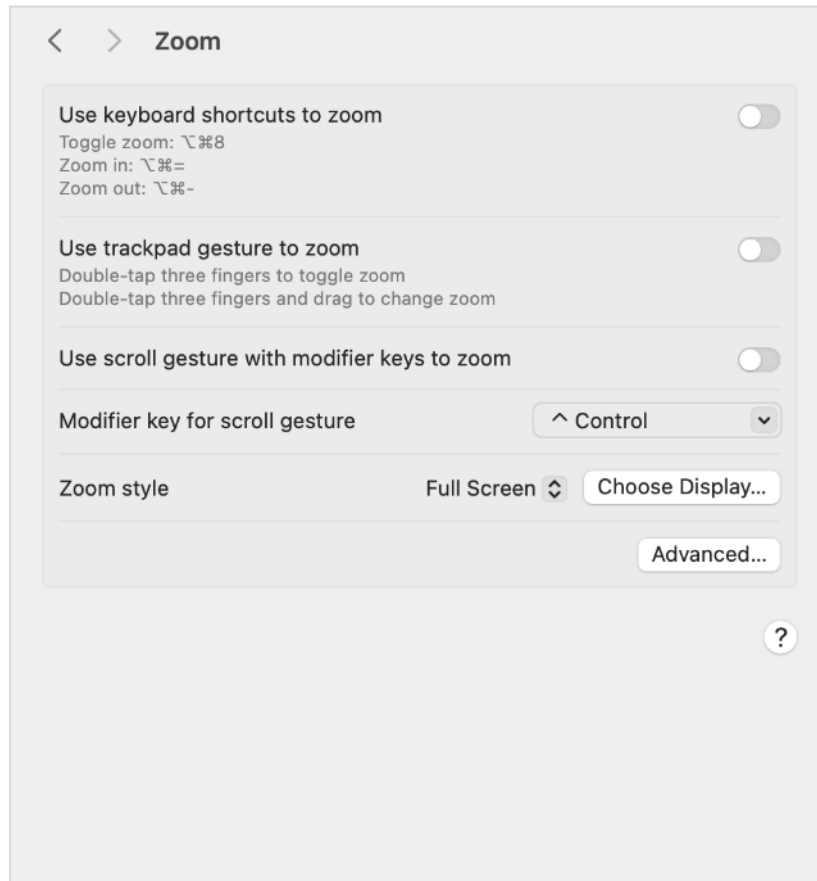


# System Settings (Slide 2 of 3)



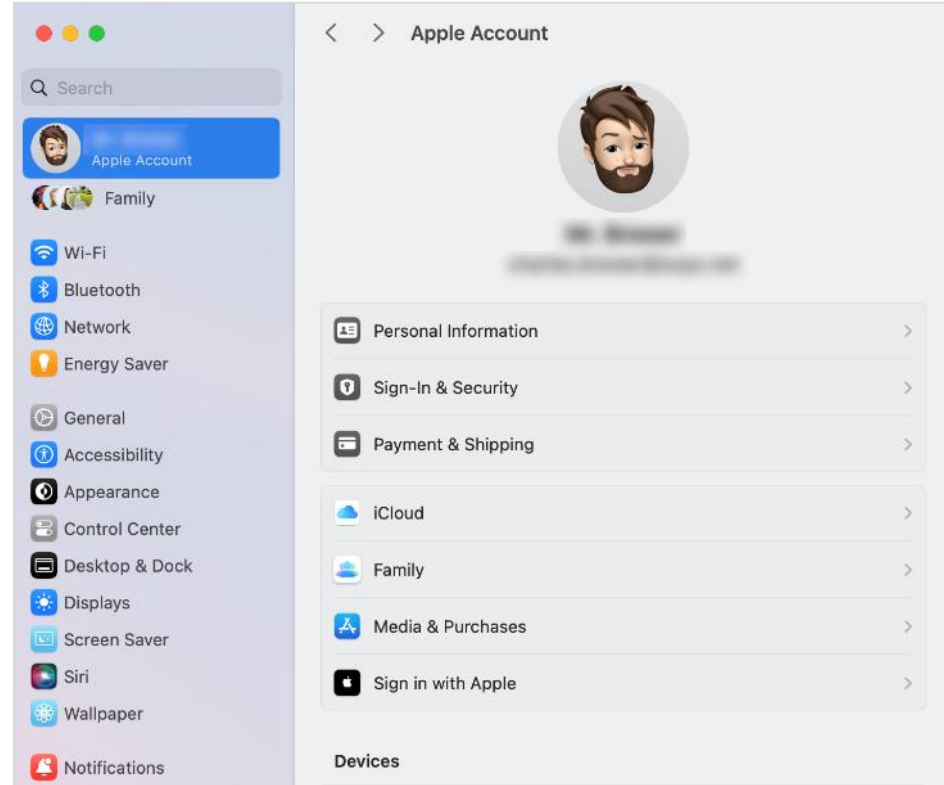
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# System Settings (Slide 3 of 3)



# Security and User Management (Slide 1 of 2)

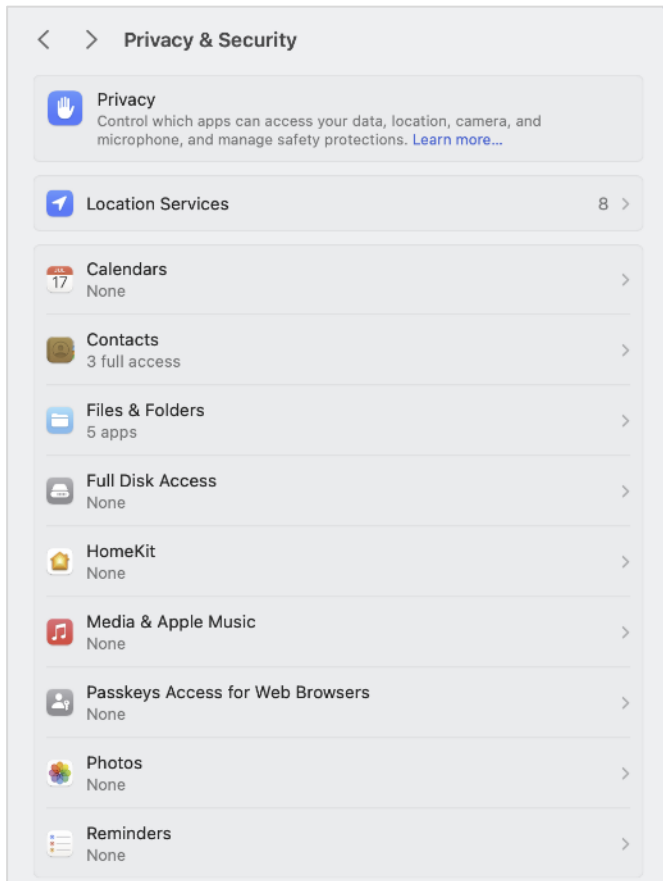
- System Settings > Users & Groups
- Apple ID
  - Appstore purchases
  - iCloud sync and data storage



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# Security and User Management (Slide 2 of 2)



*Screenshot reprinted with permission from Apple Inc.*

- Privacy and Security
  - Analytics
  - Telemetry data
  - Permissions
- Internet Accounts
  - Keychain
- File Vault
  - Disk encryption

# iCloud and Continuity (Slide 1 of 2)

- Cloud storage including Mail, Contacts, Calendar, Photos, Notes and Reminders
- FaceTime
  - Audio and video call service
- iMessage
  - Apple's proprietary messaging service
- iCloud Drive
  - Store and access files
- Continuity
  - Switch between devices seamlessly
- AirDrop
  - File sharing service direct between two Apple devices

# iCloud and Continuity (slide 2 of 2)

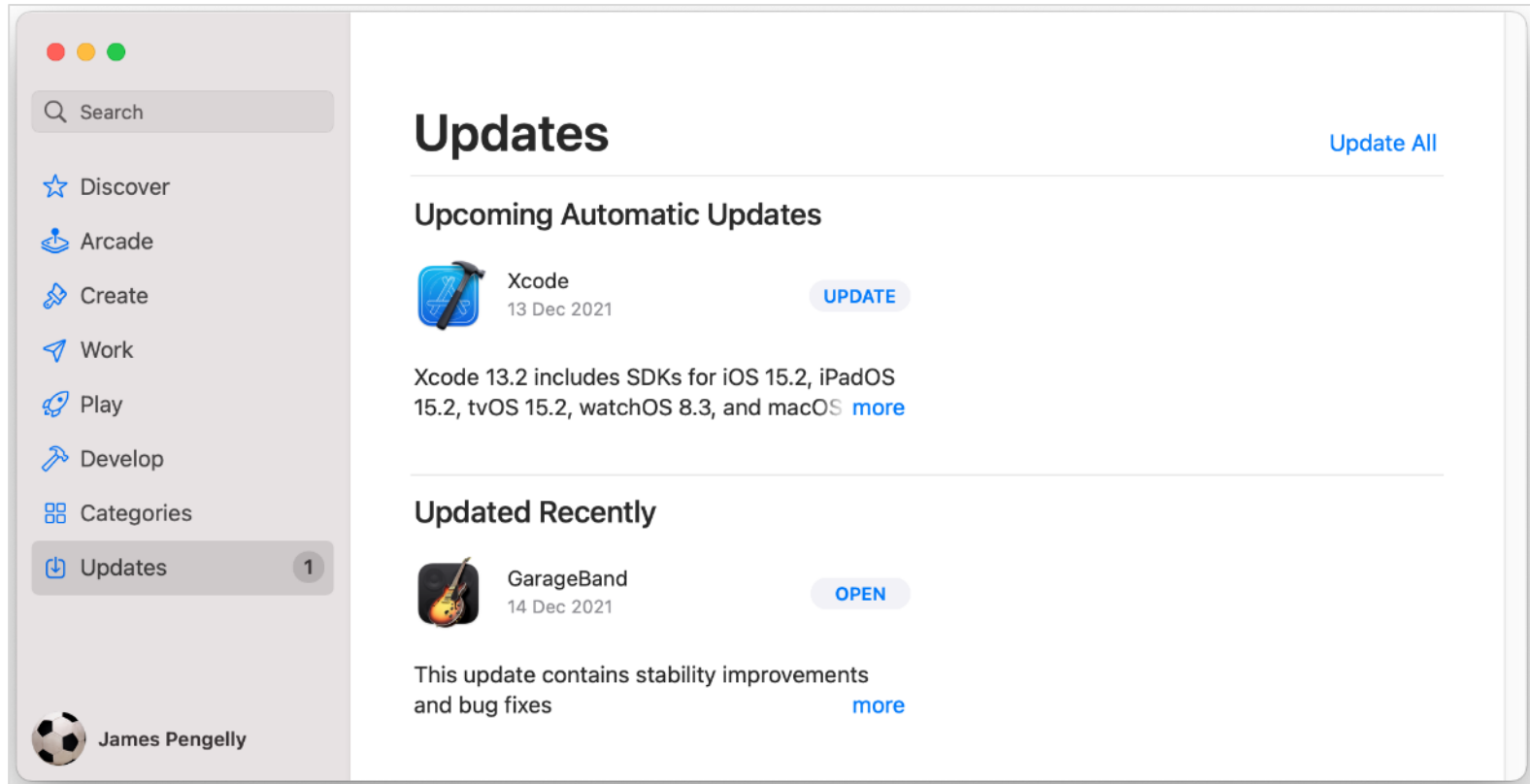


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# App Installation and Management (Slide 1 of 3)

- App Store
  - Applications and OS updates and new releases
  - Apple ID required
- System Settings > Security and Privacy to adjust settings
- .dmg
  - Disk image like Windows .iso files
- .pkg
  - Package files

# App Installation and Management (Slide 2 of 3)



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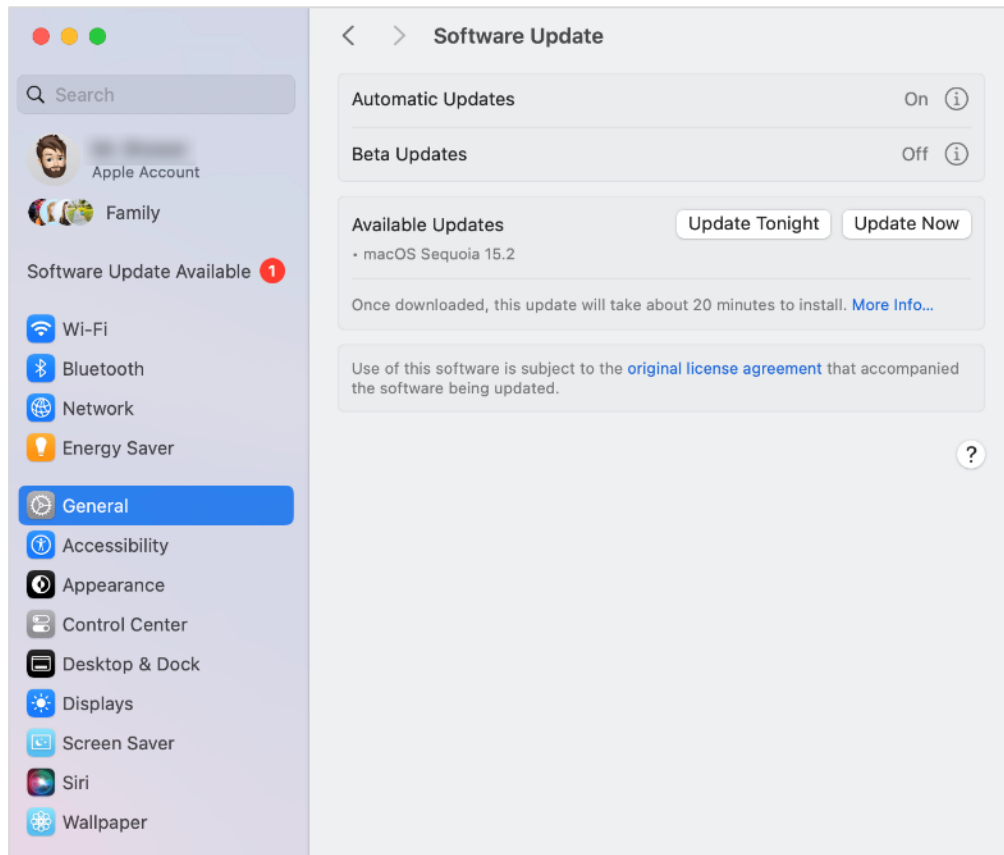
# App Installation and Management (Slide 3 of 3)

- Uninstall
  - Drag app to Trash
- Ensure regular updates are enabled for Antivirus
- Corporate Restrictions
  - Managed in Business Manager portal

# OS and App Updates (Slide 1 of 2)

- App Store
  - Enable auto download of updates
- macOS updates
  - System Settings > General > Software Update
- Rapid Security Response (RSR)
  - Security updates direct from Apple
  - Enable automatic install is best practice

# OS and App Updates (slide 2 of 2)



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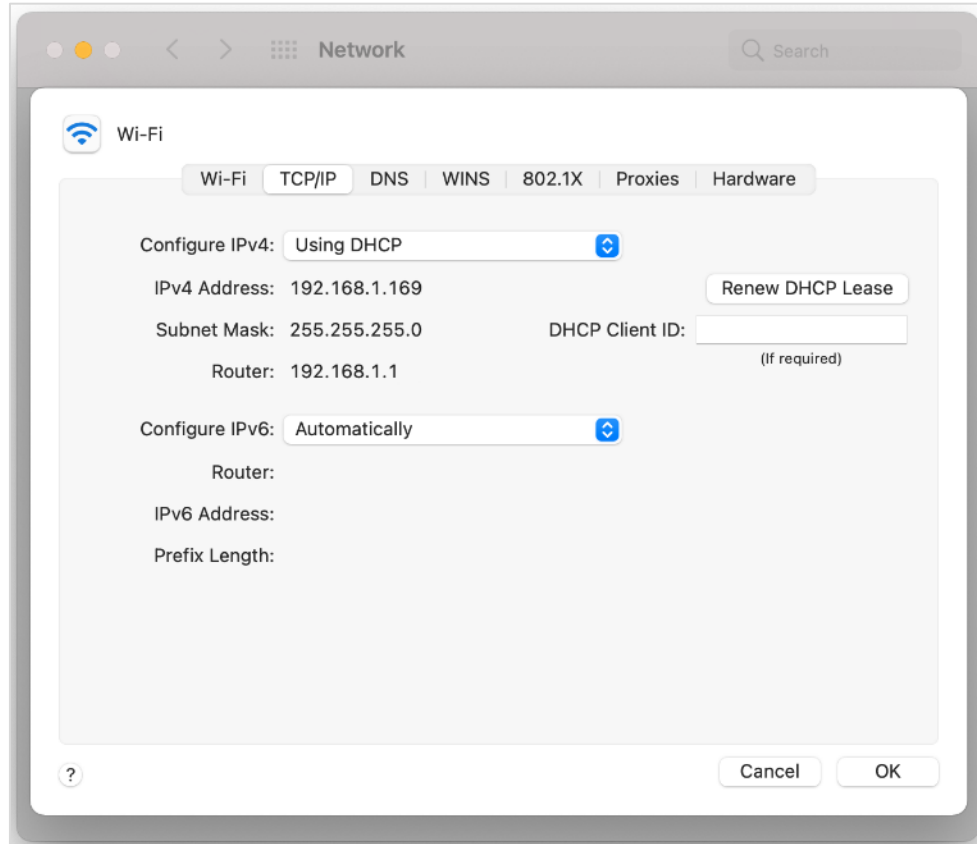
# Network and Device Settings (Slide 1 of 3)

- Status menu > Advanced
  - Configure IP, DNS, and Wi-Fi options
- Printers & Scanners
  - Add and manage devices
- Disk Utility used to verify and repair disk or file system



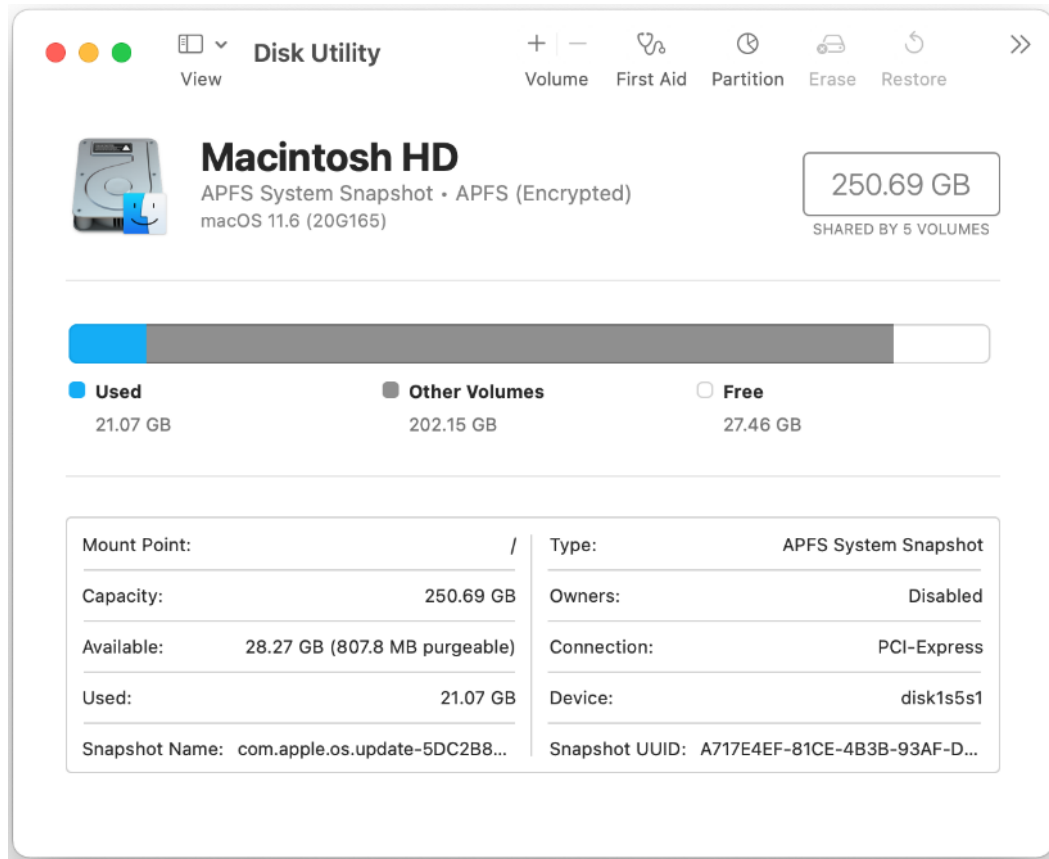
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# Network and Device Settings (Slide 2 of 3)



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# Network and Device Settings (Slide 3 of 3)

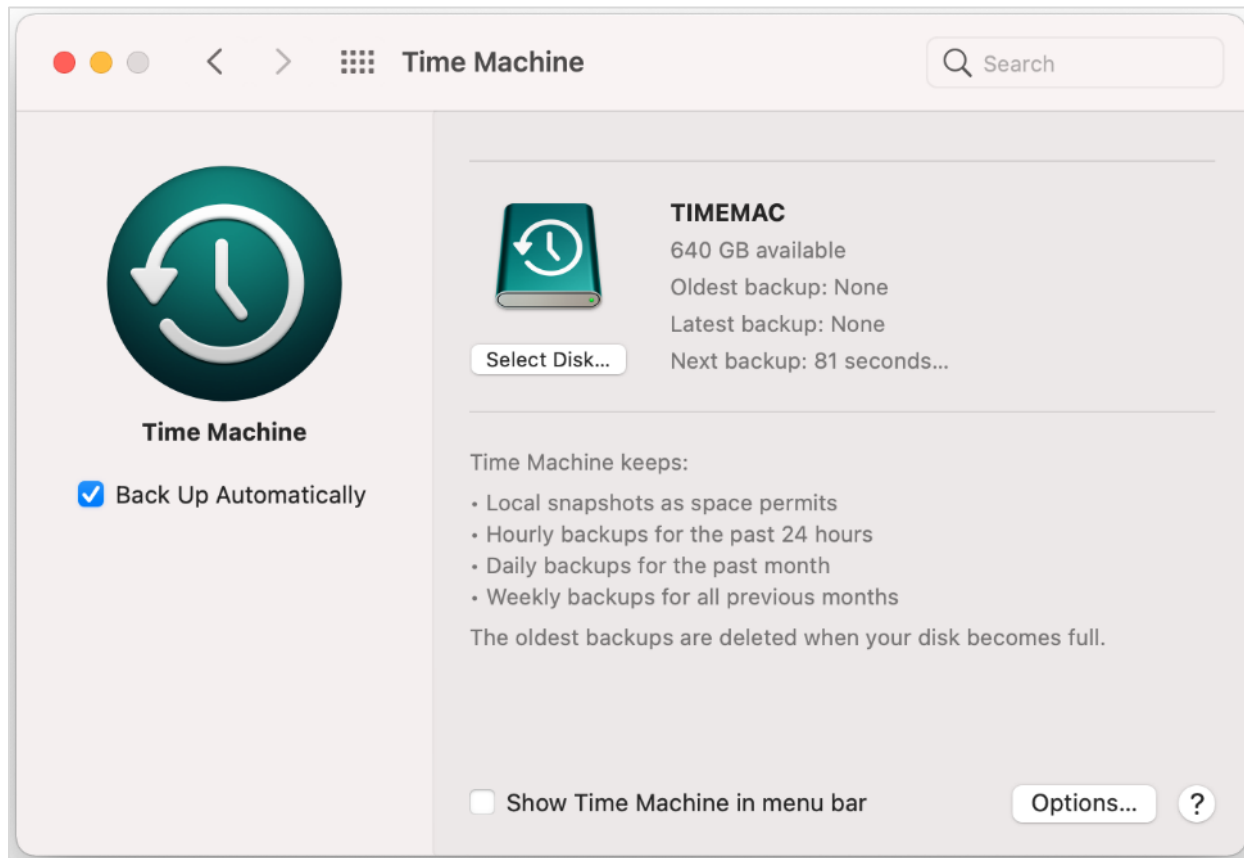


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# Time Machine Backup (Slide 1 of 2)

- Utility that uses a separate physical drive or partition to backup a system
- Partition or drive must be formatted using APFS
- Auto deletes oldest backup when full

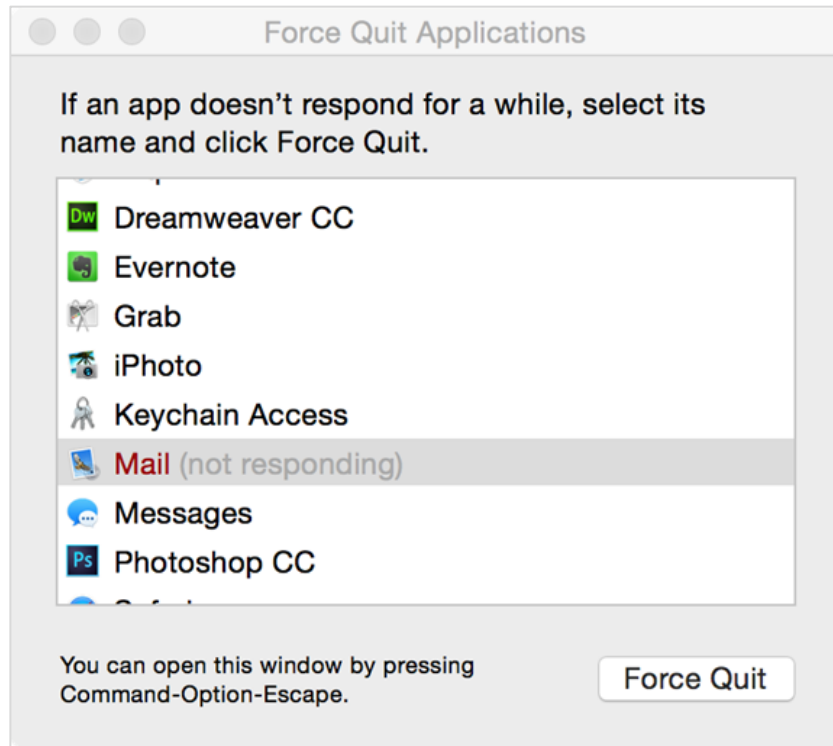
# Time Machine Backup (Slide 2 of 2)



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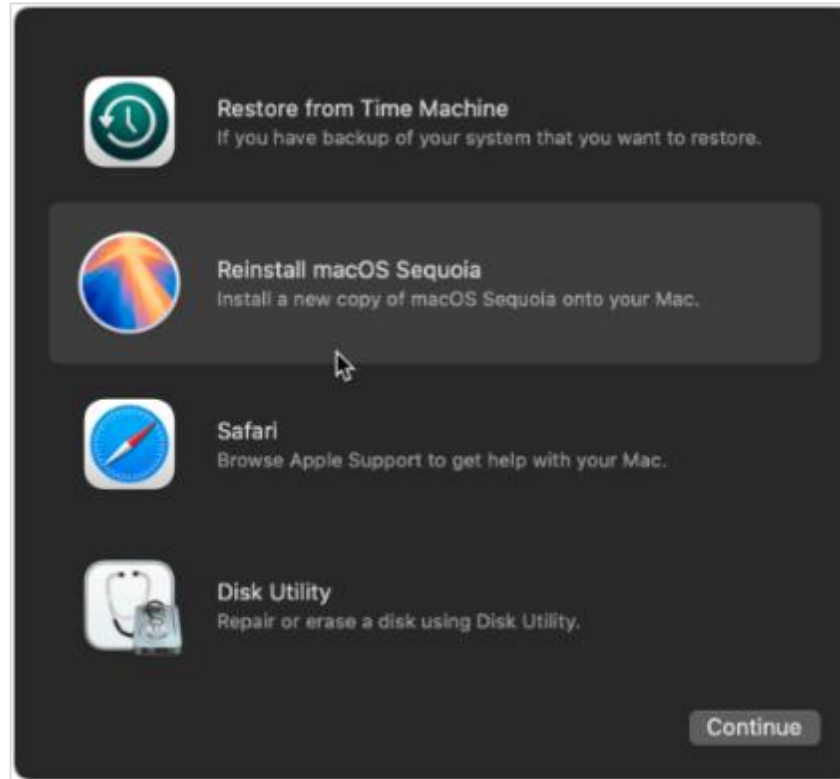
# Troubleshoot Crashes and Boot Issues (Slide 1 of 2)

- Use Force Quit from Apple menu for unresponsive apps
- Recovery Menu
  - Hold COMMAND+R while powering up Mac until Apple icon appears
- If internet access is available, Recovery can download a fresh installation of the OS



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# Troubleshoot Crashes and Boot Issues (Slide 2 of 2)



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## Lab Activity

- Challenge Live Lab: Install and Support Linux OS  
(available in CertMaster Perform and CertMaster Labs)



## Lesson Review: macOS Features

- Interface Features
- System Folders and Finder
- System Settings
- Security and User Management
- iCloud and Continuity
- App Installation and Management
- OS and App Updates
- Network and Device Settings
- Time Machine Backup
- Troubleshoot Crashes and Boot Issues